

OPERATIONAL MANAGEMENT

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Manufacturing : Input product $\rightarrow$ Process $\rightarrow$ Physical output product.

Product : all processes

Hence All manufacturing process are production.

Types of production

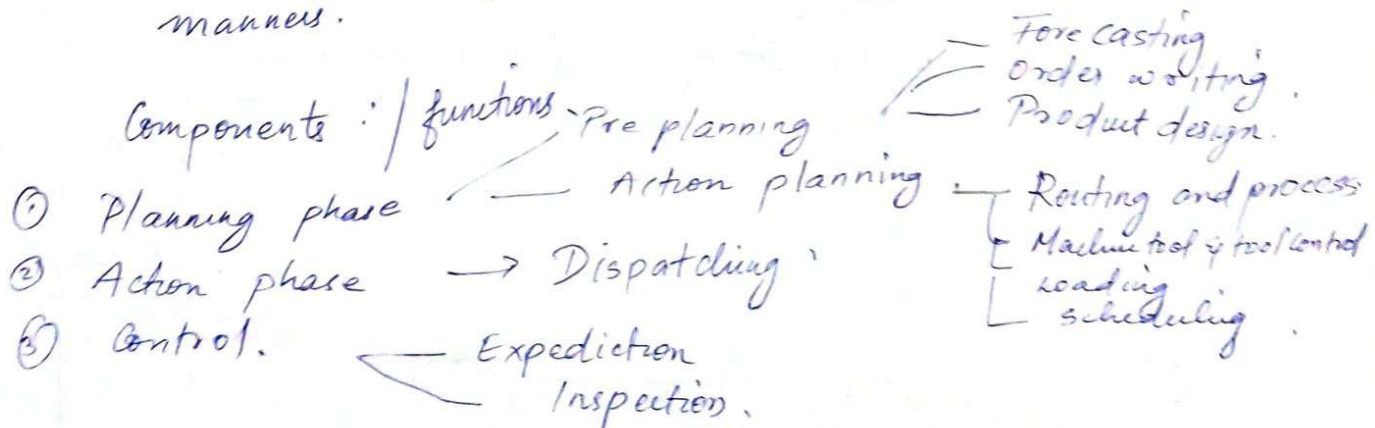
- 1) Job production - specific (single)
- 2) Mass Production - large quantity.
- 3) Batch production (Variety of Components)

Transferring techniques

- $\rightarrow$ By disintegration one input $\rightarrow$ Output 1, Output 2, Output 3. eg lathe.
- $\rightarrow$ Integration / Assembly. many input $\rightarrow$ one output.
- $\rightarrow$ Service (Note real difference b/w dp & ip)

PPC (Production planning and Control)

Direction & Coordination of firms material & physical facilities towards the attainment at pre specified goals in most efficient & economic manners.





Routing

- 1) Analysis the product.
- 2) Make or buy.
- 3) Raw material required.
- 4) Operation through which raw material undergoes.
- 5) Machines to be used.
- 6) Time required for each operations.
- 7) lot size.

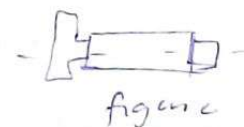
Route sheet

Part no.:

Name.:

Material.:

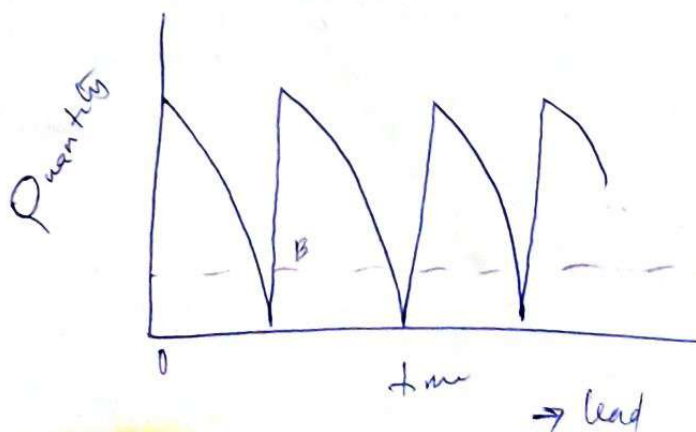
Quantity:



<u>Dept</u>	<u>Machines</u>	<u>operation</u>	<u>Description</u>	<u>tools</u>	<u>Jig/fix</u>	<u>Time.</u> <u>setup</u> <u>operat.</u>
forging	P 11/15	①	forging			

If the Dept Column is not there then it is Operation sheet.

Inventory - (storage of necessary thing - stock)



- Parts
- Tools
- Material ...

reorder.

→ lead

31/1/24

Types of Inventories parts required for final product

Direct Inventories : final products required parts
Indirect Inventories : Tools, machines, pipeline
Transport Vehicle.

{ Decoupling Inventories : Tools for maintenance.

{ Anticipation Inventories : Used for

{ Buffer Inventories : Demand change

★

Inventory Control Analysis

ABC Always Better Control

Divide Inventory based on Importance of money.

Categories → A - 75-80% (More Control)
→ B - 15-20% (Less Control)
→ C - 5-10%.

Write
Annual demand x price
in decreasing order.

No of Items	Value	
320	1500000	75% A
1000	400000	20% B
2680	100000	5% C
	<u>2000000</u>	

① for the following data of Inventory of a retail store conduct ABC Analysis

Sno	Annual demand	Price	Annual usage value
1	7200	20	144000
2	25000	30	750000
	500	65	32500
3	25000	0.5	12500
4	750	70	52500
5	100	6	600
6	120	300	36000
7	240	25	6000
	120	400	48000
8	1200	8	9600
	100	200	20000
9	36	300	10800
10	24	500	12000
11	120	200	24000
12	100	45	4500
13	48	4600	220800
			<u>1383800</u>

		Proportion	Cumulative	
		Usage	54.20	54.20
1	750000			
2	220800	15.96	70.18	A
3	144000	10.41	80.58	
4	52500	3.8	84.38	B
5	48000	3.5	87.88	
6	30000	2.6	90.48	
7	32500	2.34	92.82	
8	24000	1.7	94.52	
9	20000	1.4	95.92	C
10	12500	0.9	96.82	
11	12000	0.86	97.68	
12	10800	0.78	98.46	
13	9600	0.69	99.15	
14	6000	0.42	99.58	
15	4500	0.32	99.9	
16	600	0.043	99.943	

FSN analysis

F - fast moving
S - slow moving
N - Non moving

Based on rank of utilisation

HML Analysis

H - High
M - Medium
L - low

(Price Criteria
(unit price))

SL no	Annual Demand	Price	descending	Proportionate Value $\frac{x_i}{\sum x_i}$	cumulative
1	7200	20	7200	0.412×100	41.21 F
2	25000	30	25000	0.412×100	82.42
3	500	65	7200	0.118×100	94.22
4	25000	0.5	1200	0.019×100	96.19 S
5	750	70	750	1.2	97.42
6	100	6	500	0.824	98.244
7	120	300	240	0.39	98.634
8	940	25	120	0.19	98.824
9	120	400	120	0.19	99.014 N
10	1200	8	120	0.19	99.204
11	100	200	100	0.16	99.364
12	36	300	100	0.16	99.524
13	24	500	100	0.16	99.684
14	120	200	36	0.079	99.763
15	100	45	36	0.059	100.353
16	48	4600	24	0.039	100.392
			60658		

FSN Analysis

- Write Annual demand in Descending order.
- find Total Annual demand -
- Find proportionate Annual demand = $\frac{x_i}{\sum x_i}$
- Add one by one.
- Sort 40-80% = F

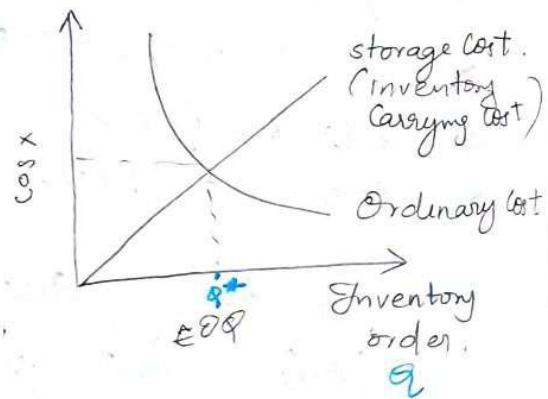
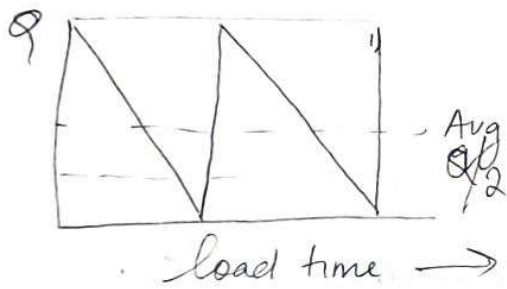
In HML Analysis (same Q.)

- Write Unit price in decreasing order.

$$\Sigma x_1 = 62469.5$$

E.O.Q. Analysis

Economic Order Quantity



Ordering cost = Storage Cost.

$$\frac{D \times C_o}{Q} = \frac{Q^2}{2} C_c$$

Q^* is present only when $\frac{D \times C_o}{Q} = \frac{Q^2}{2} C_c$

$$Q^* = \frac{2DC_o}{C_c}$$

$$Q^* = \sqrt{\frac{2DC_o}{C_c}}$$

D - Annual demand
 Q - Quantity / order

C_c - Carrying cost

C_o - Operation cost order

- ① Calculate EOQ from the following information. also calculate the no. of orders to be placed in a year. Given annual Consumption 10000 kg. Cost of placing an order = 50/- Cost per kilogram of material ₹2 Storage cost 8% of Inventory.

$$D = 10000 \text{ kg.}$$

$$C_o = 50/-$$

$$C_c = 8\% \text{ (Inventory)} = \frac{8 \times 2}{100} = 0.16$$

$$Q = \sqrt{\frac{2DC_o}{C_c}} = \frac{2 \times 10000 \times 50}{0.16} = \underline{\underline{62500 \text{ kg}}}$$

- ② A factory requires ~~one~~ 1000 1250 units of an item of material per month. each costing 30/-, The cost/order is 200 rupees. and the Inventory carrying charges 20% of Inventory. find out EOQ and no. of orders/yr

$$D = 1250 \times 12 = 15000 \text{ units. (in year)}$$

$$C_o = 200$$

$$C_c = 20\% \times 30 = 6$$

$$Q = \sqrt{\frac{2DC_o}{C_c}} = \sqrt{\frac{2 \times 15000 \times 200}{6}}$$

$$= \underline{\underline{1000}}$$

$$\text{No of orders in a year} = \frac{15000}{1000} = \underline{\underline{15}}$$

- ③ A company supplies plastic crockery to fast food restaurants in a city. One of its products is a special bowl. The bowls are sold in pack of 10 pieces at a price of ₹50 / pack. The demand for plastic bowls has been forecasted at a fairly steady rate of 40000 packs every year. The company purchased the bowl directly from the manufacturer @ 40 ₹ / pack. The ordering cost is ₹8 / order & storage cost is 10% per annum of inventory cost. Calculate economic order quantity, no. of orders needed / yr. and the total cost at EOQ level.

$$D = 40000$$

$$C_o = 8$$

$$C_c = \frac{10}{100} \times 40 = 4$$

$$Q = \sqrt{\frac{2DC_o}{C_c}} = \sqrt{\frac{2 \times 40000 \times 8}{4}}$$

$$= 400$$

$$\text{No of order} = \frac{40000}{400} = 100$$

$$\begin{aligned} \text{The total cost at EOQ level} &= 100 \times 8 \\ &= 800 + 800 \end{aligned}$$

- ④ The following information relating to a type of raw material is available annual demand 2000 units. Unit price 20 ₹ ordering cost / order = ₹ 20/- storage cost 2% of inventory & interest rate 8% per annum. Calculate EOQ and how often should be place an order in days.

$$D = 2000$$

$$C_o = 20$$

$$C_c = (8\% + 2\%) \times 20 = 2$$

$$Q = 200$$

$$= 36.5$$

- ③ A company produces product M which has a quarterly demand of 8000 units. The product requires 3kg quantity of material for every finished unit of product. Given cost of material ₹ 20/kg. Ordering cost 1000 ₹ per order and carrying cost 15% / inventory. Calculate EOQ and no. of orders to be placed in the year.

Store ledger.

maintained by the cost accounting department.

Advantages.

- Works as an internal control system.
- Tell quantity as well as value of materials.
- facilitate internal check system.
- Format of stores ledger (see ppt)
- Material request sheet. (Material requisition sheet)
 - Purpose
 - To pick items from stock
- Bin Card
 - Difference b/w Bin Card & ledger.
- Just In Time, JIT

Plant location

- Cost & benefits
- Required for new business, expansions, or shutdown due to govt or other difficulties

1000000

Needs

- change in location of demand.
- change in availability of material
- change in availability of transport.

Factors Influencing to locations.

- Proximity to market.
- Infrastructure availability
- labour and wage
- low sp taxation

$$\text{Total Cost} = \text{Fixed Cost} + [\text{Variable Cost} \times Q]$$

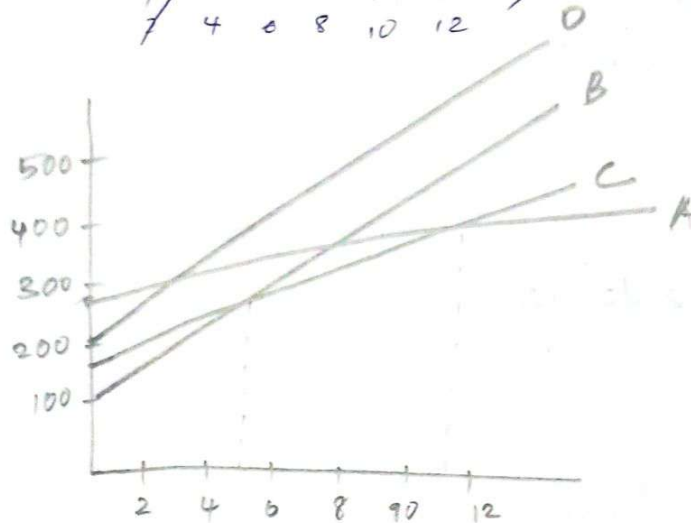
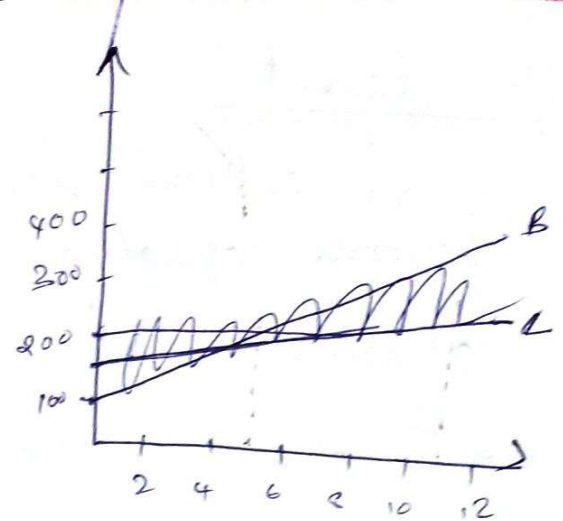
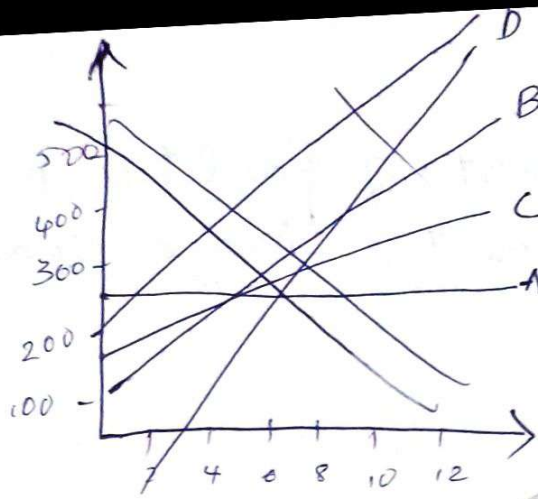
① Location	fixed cost/year	Variable cost/yr.
A	\$250000	11
B	\$100000	30
C	\$150000	20
D	\$200000	35

- Plot the total cost lines for these location on a single graph
- Identify the range of output for which each alternate is superior (lowest total cost).
- if expected output at the selected location is to be 80000 units per year which location would provide the lowest total cost.

	Q=0	Q=10000
A	250000	360000
B	100000	400000
C	150000	350000
D	200000	550000
	Total cost ↑	Total cost

$$TC = FC + VC \times Q$$

- first put $Q=0$ find TC
- Give another value for Q find TC Then plot.



(b)

The approximate ranges for which the various alternatives will yield the lowest costs are shown on the graph. Note that the location D is never superior.

(2)

An operation manager is searching for various location.			
	fixed cost	VC	Q = 0
A	150000	62	50000
B	300000	38	300000
C	500000	24	500000
D	600000	30	600000

- plot cost lines for all sites
- solve algebraically for break even analysis over relevant ranges.

★ Plant layout

Is an optimum arrangement of facilities including personal, operating equipment, storage space etc.

Objectives

- To place in a maximum utilisation position.
- facilitate easy supervision.

Principles of

- Minimum material handling.

Factors Influencing

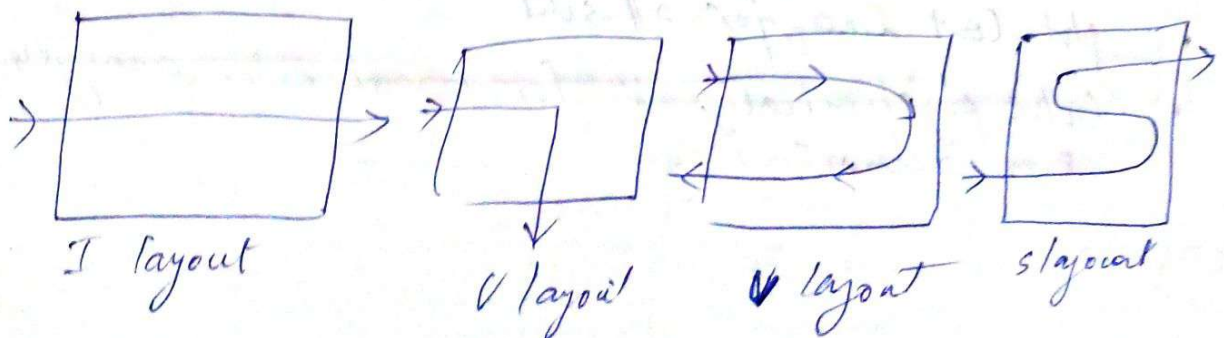
- Production type
- Scale of production
- Availability of total area

Relevance

Types of layout

- Process layout / functional layout → used in products with small batches.
- Product / line layout.
- Cellular / Group layout.

Plant layout according to flow pattern



O $\rightarrow$ Operation.

$\Rightarrow$ $\rightarrow$ Transformation transport.

D $\rightarrow$ Delay.

$\square$ $\rightarrow$ Inspection.

∇ $\rightarrow$ storage.

Travel chart

★ Material flow

★ No of movements.

★

	A	B
A		5
B	10	
C		6

Make a travel chart and simplify the travel chart to obtain optimum plant layout.

Given Movements from

A	B	C
D	E	F

$$A \rightarrow B = 20$$

$$B \rightarrow A = 10$$

$$B \rightarrow C = 15$$

$$A \rightarrow F = 25$$

$$C + D = 30$$

$$D + C = 50$$

$$D + F = 40$$

$$E \rightarrow F = 10$$

from \ to	A	B	C	D	E	F
A		10				
B	20					
C		15		50		
D			30			
E						
F	25			40	10	

	A	B	C	D	E	F
A						
B	30					
C		15				
D			80			
E						
F				40	25	

C	D	F
B	A	E

Material handling.

→ Unit load concept.

defined as number of items arranged such that they can be handled as single object

- 1) Palletisation : small open boxes.
 - 2) Unitisation : sealed
 - 3) Containerisation : In box - small items large quantity.
 - 4) Simplification
 - 5) Gravity
 - 6) Space utilisation.
 - 7) Mechanisation
 - 8) automation.
- ⑩ equipment selection
 - 11 standardisation
 - 12 Dead weight
 - 13 Utilization.
 - 14 Adaptability

- plant location
- plant layout
- Material handling
- Maintenance.

} Internal.